Spring capacitor

Y17 (2017) — English translation

A non-conducting spring with stiffness k is attached to the plates of a flat capacitor with capacity C_0. In this state, the distance between the plates L and the spring is relaxed. The plates were connected by a resistor and an external constant uniform electric field of intensity E was turned on so that the field vector is perpendicular to the plates.

A1	Find the final charge on the plates of the capacitor q.	**1.00**

A2	Find the change in the distance Δx between them, considering the charge flow process to be quasi-static.	**1.00**

A3	What amount of heat Q will be released on the resistor during this process?	**2.00**

The system is returned to its original state (without a resistor and external field). The capacitor is connected to a source of electric current with a constant voltage U.

B1	Assuming that the change in the distance between the plates $\Delta x \ll L$, find the capacitance C of the capacitor after establishing all transient processes.	**2.50**

The dependence of the capacitance C of the capacitor on the voltage U can be represented as: $C = C_0\left(1 + \alpha U^2\right)$, where α is a constant that does not depend on voltage.

C1	What is α equal to?	**0.50**

The capacitor is connected to an alternating current source, on which the voltage varies according to the law $U = U_0 \sin \omega t$. The dependence of the current strength in the circuit on time can be represented as: $I = I_0\left(A_1 \cos \omega t + A_3 \cos 3\omega t\right)$, where A_1 and A_3 are dimensionless coefficients. Due to the nonlinearity in the capacitance of the capacitor, the current in the circuit has a third harmonic $A_3 \cos 3\omega t$.

D1	Find the ratio of the amplitudes of the third and fundamental harmonics A_3/A_1. Consider that ω is much less than the frequency of natural mechanical vibrations, i.e. the delay due to mechanical inertia can be neglected. The following relationship may be useful: $4\cos^3 x = 3\cos x + \cos 3x$.	**3.00**

Source: `https://pho.rs/p/3038`